

Graph-Based Village Typology Identification

for Rural Revitalization Planning

Tsinghua University × Ji County Rural Development Project

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https://github.com/olivia3395/village_graph_analysis

Motivation & Problem Statement

Rural China faces a fundamental planning challenge: hundreds of administratively equivalent villages within a single county have vastly different development trajectories, resource endowments, and support needs. Traditional planning approaches rely on expert judgment and coarse administrative classifications that fail to capture the heterogeneity within rural communities.

This project, conducted as part of a national-level rural revitalization initiative in Ji County, Tianjin, China, addresses this gap by constructing a **graph-based machine learning system** that integrates multi-source socio-economic and spatial data to automatically identify village typologies and prioritise revitalization investment. The central research question is: *given high-dimensional, multi-source village indicators and the geographic network structure of a county, can we discover meaningful and actionable village typologies that support evidence-based planning?*

Data Sources & Multi-Source Feature Integration

Six categories of village-level indicators were fused into a unified feature representation, drawn from the following data sources:

- **Geographic** (County Planning Bureau GIS exports): GPS coordinates (longitude, latitude), elevation (m), terrain slope (degrees), and geodesic distance to the county seat (km).
- **Economic** (National Rural Statistical Yearbook): Per-capita annual income (CNY), employment rate, industry diversity index, tourism revenue (CNY), and market access score.
- **Social** (China Rural Household Survey): Total population, youth ratio (%), elderly ratio (%), education index, and healthcare access score.
- **Infrastructure** (Tianjin Infrastructure Database): Road surface quality score, broadband coverage rate, water supply reliability, and electricity reliability index.
- **Agricultural**: Cultivated land area (mu), crop diversity index, mechanisation rate, and irrigation coverage ratio.
- **Planning** (County Land-Use Registry): Current development stage, policy priority zone designation, land-use intensity, and environmental constraint level.

Preprocessing Pipeline

Raw data was processed through a four-stage pipeline:

1. **Missing value imputation**: median imputation per numeric column, applied independently within each feature group.

2. **Outlier clipping:** IQR-based clipping at $[Q_1 - 3 \cdot \text{IQR}, Q_3 + 3 \cdot \text{IQR}]$ to suppress measurement errors without discarding records.
3. **Min-max normalisation:** each feature scaled to $[0, 1]$ within its group to prevent high-magnitude indicators (e.g. income in CNY) from dominating distance computations.
4. **Composite indicator construction:** four group-level composite scores were derived as weighted averages of normalised sub-indicators:

$$\begin{aligned}
\text{EconomicVitality} &= \frac{1}{4}(\text{income} + \text{employment} + \text{market_access} + \text{industry_div}) \\
\text{SocialWelfare} &= \frac{1}{3}(\text{education} + \text{healthcare} + \text{youth_ratio}) \\
\text{InfrastructureScore} &= \frac{1}{4}(\text{road} + \text{broadband} + \text{water} + \text{electricity}) \\
\text{DevelopmentIndex} &= 0.35 \cdot \text{EconVit} + 0.25 \cdot \text{SocWel} + 0.25 \cdot \text{InfraScore} + 0.15 \cdot \text{AgriCap} \\
\text{RevitalizationNeed} &= 1 - \text{DevelopmentIndex}
\end{aligned}$$

The *RevitalizationNeed* score directly operationalises the policy goal: villages scoring near 1 have the most urgent need for external support.

Village Graph Construction

Villages were represented as nodes in a weighted undirected graph $G = (V, E, W)$, where $|V| = N$ (one node per village) and edges encode the joint spatial–socio-economic relationship between village pairs.

Edge Weight Formulation

The hybrid edge weight between villages i and j is:

$$w(i, j) = \underbrace{\alpha \cdot \exp\left(-\frac{d_{ij}}{\sigma}\right)}_{\text{spatial similarity}} + (1 - \alpha) \cdot \underbrace{\frac{x_i \cdot x_j}{\|x_i\| \|x_j\|}}_{\text{feature similarity (cosine)}},$$

where d_{ij} is the haversine geodesic distance (km), $\sigma = 7.5$ km is the spatial decay bandwidth, and $\alpha = 0.4$ balances spatial versus feature similarity. Edges below a minimum weight threshold of 0.1 are pruned to keep the graph sparse.

Rationale for the hybrid design. A purely spatial graph ($\alpha = 1$) connects geographically neighbouring villages regardless of their socio-economic similarity, causing disparate village types to cluster together simply because of physical proximity. A purely feature-based graph ($\alpha = 0$) ignores planning boundaries, road connectivity, and administrative geography, producing clusters that are uninterpretable to county planners. The hybrid captures both dimensions: two distant but equally underdeveloped villages receive a meaningful edge weight through their feature similarity, while geographically adjacent villages are also connected even if their features differ slightly.

Structural Node Features

In addition to raw socio-economic attributes, graph-structural properties were computed per node and included in the feature set:

Feature	Interpretation in Planning Context
Degree	Number of connected neighbouring villages
Weighted degree	Total connectivity strength (sum of edge weights)
Clustering coefficient	How tightly integrated into a local cluster
Betweenness centrality	Bridge villages on inter-cluster paths (hub candidates for shared services)
Closeness centrality	Ease of reach from all other villages (accessibility)
PageRank	Importance relative to the importance of neighbours
Local density	Mean edge weight in the 1-hop neighbourhood

Graph Learning & Clustering Methods

Three complementary clustering approaches were implemented and compared, each exploiting the graph structure differently.

Spectral Clustering

The normalised graph Laplacian is defined as:

$$L = I - D^{-1/2}AD^{-1/2},$$

where A is the weighted adjacency matrix and $D = \text{diag}(A\mathbf{1})$ is the degree matrix. The k eigenvectors corresponding to the k smallest eigenvalues of L form the spectral embedding matrix $U \in \mathbb{R}^{N \times k}$. After row-normalising U (projecting each village onto the unit sphere in $\mathbb{R}^k$), k -means clustering is applied to obtain final assignments.

The **Fiedler vector** (second-smallest eigenvector) provides a meaningful one-dimensional development gradient over the county. Villages ordered by this vector transition smoothly from underdeveloped to highly developed, giving planners an interpretable spatial continuum without any further analysis. The **eigengap** $\lambda_{k+1} - \lambda_k$ is monitored to validate the choice of k : a large gap indicates a well-defined cluster structure.

Louvain Community Detection

The Louvain algorithm maximises the modularity objective:

$$Q = \frac{1}{2m} \sum_{i,j} \left[A_{ij} - \frac{d_i d_j}{2m} \right] \delta(c_i, c_j),$$

where $m = \frac{1}{2} \sum_{i,j} A_{ij}$ is the total edge weight, $d_i = \sum_j A_{ij}$ is the weighted degree of node i , and $\delta(c_i, c_j) = 1$ iff villages i and j are in the same community. A tunable resolution parameter γ modulates cluster granularity: higher γ yields more, smaller communities suitable for township-level planning, while lower γ produces fewer, larger communities matching county-level administrative divisions.

Node2Vec Embedding + K-Means

Node2Vec generates node embeddings by simulating biased second-order random walks on the village graph, controlled by two parameters:

- p (return parameter): probability of returning to the previously visited node. Low p encourages local exploration (BFS-like).
- q (in-out parameter): probability of moving to a node further from the source. Low q encourages depth-first exploration, capturing global structural roles.

We use $q = 0.5 < 1$ (DFS-biased) so that structurally equivalent villages (e.g. two isolated peripheral villages in different parts of the county) obtain similar embeddings and cluster together, even when geographically separated. Random walk co-occurrences are weighted by PPMI (Positive Pointwise Mutual Information) and compressed via truncated SVD to yield $\mathbb{R}^{32}$ embeddings, then clustered with k -means.

Village Typology Identification

Each cluster was assigned a canonical typology label via a **greedy archetype-signature matching** procedure. For each cluster, the mean value of key indicator features is compared against domain-defined directional signatures (e.g. “Agricultural Core” expects high *cultivated_area*, high *mechanisation_rate*, low *tourism_revenue*). A score is computed per (cluster, typology) pair and the highest-scoring unassigned pair is greedily matched until all clusters are labelled.

Typology	Key Distinguishing Features	Priority
Agricultural Core	High cultivated area, mechanisation, irrigation	Moderate
Tourism & Cultural	High tourism revenue, elevation, industry diversity	High
Peri-urban Transition	Low dist. to county, high road quality, broadband	Low
Underdeveloped Priority	Low income, high elderly ratio, poor infra.	Highest
Industrial & Commercial	High income, industry diversity, market access	Low

Development Prioritisation

Each village receives a **growth potential score**:

$$\text{GrowthPotential}(v) = \text{ResourceScore}(v) \times \exp\left(-2 \left| \text{DevIndex}(v) - 0.4 \right| \right),$$

which peaks for villages at moderate development (≈ 0.4) with strong natural or economic resources — those with the highest marginal return on investment. Villages are then assigned to three revitalization priority tiers:

- **Tier 1 — Urgent:** RevitalizationNeed > 0.65 . Direct fiscal transfer and infrastructure investment required.
- **Tier 2 — Moderate:** $0.40 < \text{RevitalizationNeed} \leq 0.65$. Targeted sector support (tourism development, agricultural modernisation).
- **Tier 3 — Self-sufficient:** RevitalizationNeed ≤ 0.40 . Minimal intervention; monitor for negative transitions.

Evaluation

Clustering quality was assessed using both internal and external metrics.

Internal metrics (no ground truth required):

- *Silhouette coefficient*: measures intra-cluster cohesion vs. inter-cluster separation; range $[-1, +1]$, higher is better.
- *Calinski-Harabasz index*: ratio of between- to within-cluster variance; higher is better.
- *Davies-Bouldin score*: mean cluster similarity (lower is better).

External metrics (against ground-truth typology labels from synthetic data):

- *Adjusted Rand Index (ARI)*: chance-corrected agreement between predicted and true partitions; range $[-1, +1]$.
- *Normalised Mutual Information (NMI)*: information overlap between partitions; range $[0, 1]$.
- *V-measure*: harmonic mean of homogeneity and completeness.

Method	ARI	NMI	V-measure	Silhouette	Davies-Bouldin
Spectral Clustering	0.782	0.824	0.831	0.431	0.812
Louvain	0.694	0.782	0.791	0.399	0.934
Node2Vec + K-Means	0.651	0.723	0.731	0.374	1.021

Spectral clustering consistently outperforms the alternatives by leveraging the full graph Laplacian structure, which encodes both spatial proximity and feature similarity into a single coherent representation. The eigengap at $k = 5$ confirms that five clusters is a structurally justified choice for the Ji County village network.

Key Contributions

1. Designed and implemented an end-to-end multi-source data integration and graph construction pipeline for rural village analysis, handling heterogeneous data from six administrative and survey sources.
2. Proposed a hybrid spatial–feature edge weighting scheme that captures both geographic realities and socio-economic similarity, producing a graph structure more suitable for planning than either purely spatial or purely feature-based alternatives.
3. Implemented and systematically compared three graph-based clustering methods (spectral, Louvain, Node2Vec), demonstrating the superiority of spectral clustering for this task and providing planners with the interpretable Fiedler development gradient as a by-product.
4. Developed a domain-grounded typology labelling system and a three-tier revitalization priority framework that directly translates clustering outputs into actionable planning recommendations, supporting evidence-based allocation of national revitalization funds.